Akhila Anya Pisupati 225 homework 2 1/25/26

Problem 1.16 $P_{+} = \frac{1}{2}$ $P_{+x} = \frac{3}{4}$ $P_{+y} = 0.067$

$P_{-} = \frac{1}{2}$ $P_{-x} = \frac{1}{4}$ $P_{-y} = 0.933$

$P_{+} = |\langle + | \psi_{in} \rangle|^2 = \frac{1}{2} = P_{-} = |\langle - | \psi_{in} \rangle|^2$ $|\psi_{in}\rangle = a|+\rangle + b|-\rangle$

$|\langle + | [a|+\rangle + b|-\rangle]|^2 = |\langle - | [a|+\rangle + b|-\rangle]|^2 = \frac{1}{2}$

$|a\langle + | + \rangle + b\langle + | - \rangle|^2 = |a\langle - | + \rangle + b\langle - | - \rangle|^2 = \frac{1}{2}$

$|a|^2 = |b|^2 = \frac{1}{2} \Rightarrow a = b = \frac{1}{\sqrt{2}}$

$|\psi_{in}\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle = |+_x\rangle$

$\boxed{\text{Problem 2}}$ A. Pisupati

$|\psi\rangle = \frac{1}{2}|+\rangle - \frac{\sqrt{3}}{2}|-\rangle = \frac{1}{2}\left[|+\rangle - \sqrt{3}|-\rangle\right]$

$|+_y\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}i|-\rangle$, so $\langle +_y| = \frac{1}{\sqrt{2}}\left[\langle +| - i\langle -|\right]$

$P_{+y} = |\langle +_y|\psi\rangle|^2 = \left|\frac{1}{\sqrt{2}}[\langle +| - i\langle -|]\left[\frac{1}{2}|+\rangle - \frac{\sqrt{3}}{2}|-\rangle\right]\right|^2$

$= \left|\frac{1}{\sqrt{2}}\left[\frac{1}{2}\langle +|+\rangle - \frac{\sqrt{3}}{2}\langle +|-\rangle - \frac{1}{2}i\langle -|+\rangle + \frac{\sqrt{3}}{2}i\langle -|-\rangle\right]\right|^2$

$= \frac{1}{2}\left|\frac{1}{2} + \frac{\sqrt{3}}{2}i\right|^2 = \frac{1}{2}\left(\sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}\right)^2 = \frac{1}{2}\left(\sqrt{\frac{1}{4} + \frac{3}{4}}\right)^2 = \frac{1}{2}\cdot 1 = \frac{1}{2}$

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$\boxed{\text{Problem 2.3}}$ Matrix representation of S_z in the S_x basis

$S_z \doteq \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in the S_x basis

$|+\rangle = \frac{1}{\sqrt{2}}|+_x\rangle + \frac{1}{\sqrt{2}}|-_x\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$|-\rangle = \frac{1}{\sqrt{2}}|+x\rangle - \frac{1}{\sqrt{2}}|-_x\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Matrix representation: $\hat{O}|\psi\rangle = \lambda|\psi\rangle$

$|+\rangle: \cancel{\frac{1}{\sqrt{2}}}\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \cancel{\frac{1}{\sqrt{2}}}\frac{\hbar}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} a+b \\ c+d \end{pmatrix} = \begin{pmatrix} \hbar/2 \\ \hbar/2 \end{pmatrix}$

$|-\rangle: \cancel{\frac{1}{\sqrt{2}}}\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \cancel{\frac{1}{\sqrt{2}}}\left(-\frac{\hbar}{2}\right)\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} a & -b \\ c & -d \end{pmatrix} = \begin{pmatrix} -\hbar/2 \\ \hbar/2 \end{pmatrix}$

$\begin{cases} a+b = \hbar/2 \Rightarrow b = \hbar/2 - a = \hbar/2 \\ a-b = -\hbar/2 = a - (\hbar/2 - a) \Rightarrow 2a = 0 \text{ so } a = 0 \end{cases}$

$\begin{cases} c+d = \hbar/2 \Rightarrow c = \hbar/2 - d = \hbar/2 \\ c-d = \hbar/2 = \frac{\hbar}{2} - d - d \Rightarrow -2d = 0 \text{ so } d = 0 \end{cases}$

So, S_z in the S_x basis $\doteq \begin{pmatrix} a & b \\ c & d \end{pmatrix} \doteq \begin{pmatrix} 0 & \hbar/2 \\ \hbar/2 & 0 \end{pmatrix}$

Diagonalization: $\det(S_z - \lambda I) = \begin{vmatrix} -\lambda & \hbar/2 \\ \hbar/2 & -\lambda \end{vmatrix} = \lambda^2 - \frac{\hbar^2}{4} = 0 \Rightarrow \lambda = \pm\frac{\hbar}{2}$ eigenvalues

Eigenvectors:

$|+\rangle: \begin{pmatrix} 0 & \hbar/2 \\ \hbar/2 & 0 \end{pmatrix}\begin{pmatrix} m \\ n \end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix} m \\ n \end{pmatrix} = \begin{pmatrix} (\hbar/2)m \\ (\hbar/2)n \end{pmatrix} \Rightarrow m = n \Rightarrow \begin{pmatrix} m \\ n \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$|-\rangle: \begin{pmatrix} 0 & \hbar/2 \\ \hbar/2 & 0 \end{pmatrix}\begin{pmatrix} p \\ q \end{pmatrix} = -\frac{\hbar}{2}\begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} (\hbar/2)p \\ (\hbar/2)q \end{pmatrix} \Rightarrow p = -q \Rightarrow \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

* eigenvectors

Verification:*

$\begin{pmatrix} 0 & \hbar/2 \\ \hbar/2 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \hbar/2 \\ \hbar/2 \end{pmatrix} \checkmark$

$\begin{pmatrix} 0 & \hbar/2 \\ \hbar/2 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} \hbar/2 \\ -\hbar/2 \end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix} 1 \\ -1 \end{pmatrix} \checkmark$

*Add a factor of $\frac{1}{\sqrt{2}}$ that cancels out anyway

$\boxed{\text{Problem 4}}$ $\hat{A}^{\dagger} = A$ and $\hat{B}^{\dagger} = B$

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(a) Is $\hat{A}^2$ Hermitian?

$$\hat{A}^2 = (\hat{A}^{\dagger})^2 = \hat{A}^{\dagger}\hat{A}^{\dagger} = \hat{A}^{\dagger 2} \rightarrow \text{Yes (?)}$$

(b) Is $\hat{A}\hat{B}$ Hermitian?

$$\hat{A}\hat{B} = \hat{A}^{\dagger}\hat{B}^{\dagger} = (\hat{B}\hat{A})^{\dagger} = (\hat{B}^{\dagger}\hat{A}^{\dagger})^{\dagger} \text{ (since, again, } \hat{A} = \hat{A}^{\dagger}; \hat{B} = \hat{B}^{\dagger})$$

$$= \hat{A}^{\dagger\dagger}\hat{B}^{\dagger\dagger} = \hat{A}\hat{B}$$

Yes, $\hat{A}\hat{B}$ is Hermitian

(c) Is $\hat{A}\hat{B} + \hat{B}\hat{A}$ Hermitian?

$$\hat{A}\hat{B} + \hat{B}\hat{A} = \hat{A}^{\dagger}\hat{B}^{\dagger} + \hat{B}^{\dagger}\hat{A}^{\dagger} = (\hat{B}\hat{A} + \hat{A}\hat{B})^{\dagger} = (\hat{B}^{\dagger}\hat{A}^{\dagger} + \hat{A}^{\dagger}\hat{B}^{\dagger})^{\dagger}$$

$$= \hat{A}^{\dagger\dagger}\hat{B}^{\dagger\dagger} + \hat{B}^{\dagger\dagger}\hat{A}^{\dagger\dagger} = \hat{A}\hat{B} + \hat{B}\hat{A}$$

It's also worthy to note that $\hat{A}^{\dagger\dagger} = \hat{A}$. Yes, this is Hermitian

(d) $\hat{A}$ must have real eigenvalues because they represent a physical, measurable outcome/observable, like the magnitude of spin in the +z direction, which is treated as an angular momentum: $S_z = +\hbar/2$. Thus, it makes no sense for a "real (lol) world" value to be complex.

Change of basis

Suppose an electron is described by the state vector $|\psi\rangle = \frac{1}{2}|+\rangle + \frac{\sqrt{3}}{2}|-\rangle$, where $|+\rangle$ and $|-\rangle$ are the basis states corresponding to spin-up and spin-down, respectively, in the *z*-direction.

A. Calculate the inner product $\langle +|\psi\rangle$.

$$\langle +|\psi\rangle = \langle +|\left[\frac{1}{2}|+\rangle + \frac{\sqrt{3}}{2}|-\rangle\right] = \frac{1}{2}\langle +|+\rangle + \frac{\sqrt{3}}{2}\langle +|-\rangle$$
$$= \frac{1}{2}(1) + \frac{\sqrt{3}}{2}(0) = \frac{1}{2}$$

B. Is this state vector normalized? Explain by discussing the probabilities of measuring spin up or down in the *z*-direction, as done in the Worksheet 2.

Yes, because $|\psi\rangle$ is normalized if its probability amplitudes (a^2 and b^2 in $|\psi\rangle = a|+\rangle + b|-\rangle$) sum to 1. In this case, $a = \frac{1}{2}$ and $b = \frac{\sqrt{3}}{2}$.

$a^2 = (\frac{1}{2})^2 = \frac{1}{4}$ and $b^2 = (\frac{\sqrt{3}}{2})^2 = \frac{3}{4}$ $\quad \frac{1}{4} + \frac{3}{4} = a^2 + b^2 = 1$

C. *Predict* (without performing explicit calculations) whether the magnitude of the inner product ${}_x\langle +|\psi\rangle$ is *greater than, less than,* or *equal to* the magnitude of the inner product $\langle +|\psi\rangle$, where $|+\rangle_x$ is the basis state corresponding to spin-up in the *x*-direction. Explain.

Greater than: finding $|+_x\rangle$ has extra nonzero coefficients and a $|-\rangle$ component so there'll be an extra additive term when finding $\langle +_x|\psi\rangle$ that makes the inner product larger than $\langle +|\psi\rangle$ alone.

D. Consider the dialogue below between three students who are attempting to determine the inner product ${}_x\langle +|\psi\rangle$.

Student 1: "This is just like the first inner product: ${}_x\langle +|-\rangle$ is zero because the plus and minus states are always orthogonal to each other. Therefore, ${}_x\langle +|\psi\rangle$ is 1/2."

Student 2: "The *x*- and *z*-directions are orthogonal to each other, so $|+\rangle_x$ is orthogonal to both $|+\rangle$ and $|-\rangle$. That means the answer is zero."

Student 3: "I don't think the inner product of ${}_x\langle +|+\rangle$ is zero. If we send an electron prepared in the state $|+\rangle$ through a S-G apparatus with magnetic field in the *x*-direction, we would have a 50/50 chance of measuring spin up and spin down. Thus, the inner product can't be zero."

Which student(s), if any, do you agree with? Explain your reasoning.

Student 3: None of their discussion is related to $\langle +_x|\psi\rangle$ but they have the right logic. Probability is the squared absolute magnitude of the inner product, so to get a nonzero probability, $\langle +_x|+\rangle$ has to be nonzero. Student 1 doesn't account for the change in direction ($\langle +_x| \neq \langle +|$) and student 2 is right that $\langle +_x|$ is orthogonal to $\langle +|$ and $\langle -|$, but not on the calculations: coefficients and multiplying components!!

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E. Determine the inner product, ${}_x\langle +|\psi\rangle$, using Dirac notation. Clearly show each step in your calculation.

$|+_x\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle$, so $\langle +_x| = \frac{1}{\sqrt{2}}\langle +| + \frac{1}{\sqrt{2}}\langle -|$

$$\langle +_x|\psi\rangle = \left[\frac{1}{\sqrt{2}}\langle +| + \frac{1}{\sqrt{2}}\langle -|\right]\left[\frac{1}{2}|+\rangle + \frac{\sqrt{3}}{2}|-\rangle\right]$$

$$= \frac{1}{2\sqrt{2}}\langle +|+\rangle + \cancel{\frac{\sqrt{3}}{2\sqrt{2}}\langle +|-\rangle} + \cancel{\frac{1}{2\sqrt{2}}\langle -|+\rangle} + \frac{\sqrt{3}}{2\sqrt{2}}\langle -|-\rangle$$

$$= \frac{1+\sqrt{3}}{2\sqrt{2}} \boxed{\approx 0.97} >> \frac{1}{2} = \langle +|\psi\rangle$$

Resolve any inconsistencies with your prediction in question C.

F. Is it possible to rewrite $|\psi\rangle$ in terms of the basis state vectors $|+\rangle_x$ and $|-\rangle_x$? Explain why or why not. Yes, because you can write the basis vectors of $|\psi\rangle$, $|+\rangle$ and $|-\rangle$, entirely with respect to $|+_x\rangle$ and $|-_x\rangle$, not the other current basis vector. It stands to reason that some multiplication would yield $|\psi\rangle$ in terms only of $|+_x\rangle$ and $|-_x\rangle$!

If it is possible, do you expect the coefficients in terms of $|+\rangle_x$ and $|-\rangle_x$ to be the same as the coefficients in terms of $|+\rangle$ and $|-\rangle$? Explain. No, because the coefficients represent projections of $|\psi\rangle$ along each respective basis ket. The state is not equally projected along the 'x' and 'z' directions, or their probabilities would be equal, which isn't happening here: $|\langle +|\psi\rangle|^2 = \frac{1}{4} \neq |\langle +_x|\psi\rangle|^2 = 0.97^2$

If it is possible to write $|\psi\rangle$ in terms of $|+\rangle_x$ and $|-\rangle_x$, do so. Clearly show your work.

$$|+\rangle = \frac{1}{\sqrt{2}}|+_x\rangle + \frac{1}{\sqrt{2}}|-_x\rangle$$

$$|-\rangle = \frac{1}{\sqrt{2}}|+_x\rangle - \frac{1}{\sqrt{2}}|-_x\rangle$$

$$|\psi\rangle = \frac{1}{2}|+\rangle + \frac{\sqrt{3}}{2}|-\rangle = \frac{1}{2\sqrt{2}}|+_x\rangle + \frac{1}{2\sqrt{2}}|-_x\rangle + \frac{\sqrt{3}}{2\sqrt{2}}|+_x\rangle - \frac{\sqrt{3}}{2\sqrt{2}}|-_x\rangle$$

$$\frac{1}{2}|+\rangle = \frac{1}{2}\left[\frac{1}{\sqrt{2}}(|+_x\rangle + |-_x\rangle)\right] = \frac{1}{2\sqrt{2}}|+_x\rangle + \frac{1}{2\sqrt{2}}|-_x\rangle$$

$$\frac{\sqrt{3}}{2}|-\rangle = \frac{\sqrt{3}}{2}\left[\frac{1}{\sqrt{2}}(|+_x\rangle - |-_x\rangle)\right] = \frac{\sqrt{3}}{2\sqrt{2}}|+_x\rangle - \frac{\sqrt{3}}{2\sqrt{2}}|-_x\rangle$$

$$|\psi\rangle = \frac{1+\sqrt{3}}{2\sqrt{2}}|+_x\rangle + \frac{1-\sqrt{3}}{2\sqrt{2}}|-_x\rangle$$